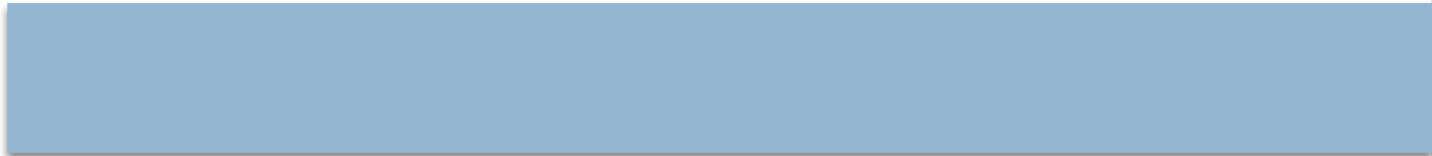


COMPUTER CASES AND SAFETY



Computer case

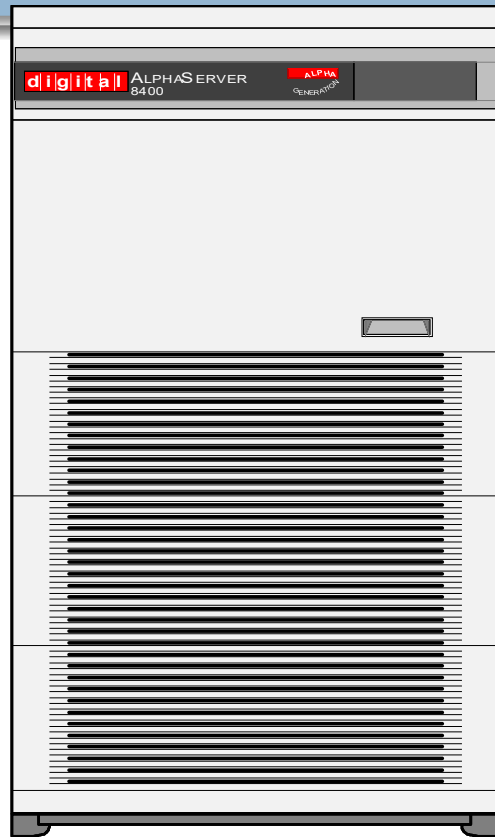
- is an enclosure that holds all the primary electronics of the PC. It is also called chassis. Case is usually made of metal.
- The real value of a case does not lie in the label, color, or how pretty it looks. Computer cases have the purpose of:
 - house all the internal components
 - offers access to the external world through ports and connectors
 - prevent delicate circuits from damage and EMI
 - prevent EMI from pc from affecting surrounding devices such as TV, telephone, etc

Computer cases ...

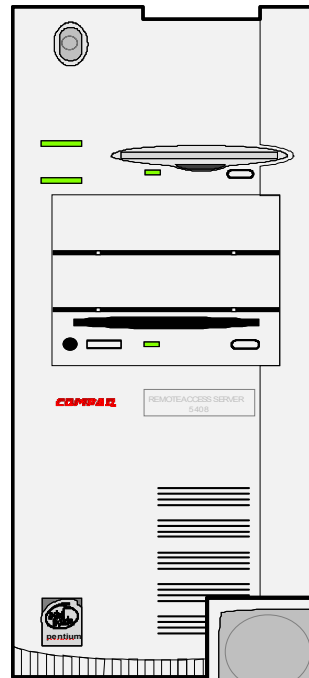
- EMI-is a radio frequency that is emitted from electronic or electrical device. It interferes with surrounding equipment.
- A computer interferes with radio, telephone, TV, etc when it generates EMI. Running a computer without cases causes interference to surrounding devices.
- It is *not* a good idea to run a computer for extended periods of time with the case open or removed entirely.
- This not only produces EMI, but also results in improper airflow and reduced cooling of the system components.

Types of cases

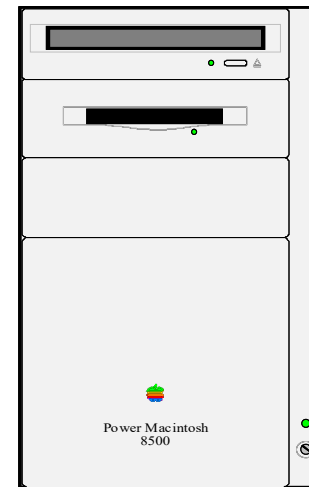
- There are different types of cases:
 - ▣ Full-sized tower
 - ▣ Mid-size tower
 - ▣ Mini-tower
 - ▣ Desktop
 - ▣ Slim desktop
- Slim desktops integrate the display into housing. The monitor and system unit are combined into one unit. E.g. imac G5/



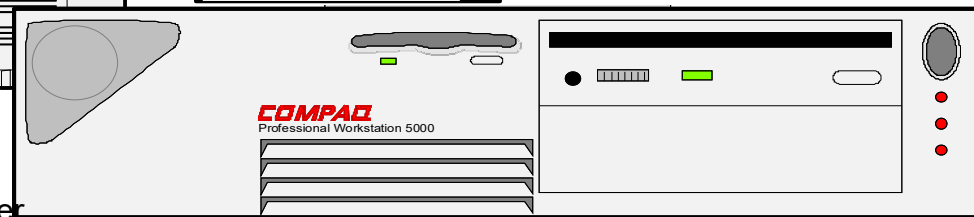
Full-size tower



Mid-size tower



Mini-tower



Desktop



micro, mini, mid, full tower, & desktop cases

- 
- When we consider a case, the following things are cared about:
 - ▣ The bigger the box, the more components it can hold. This gives it more expansion capability, better air flow which is good for cooling. Large cases are easier to work with.
 - ▣ The more compact the box, the less expansion potential it has. Working with such cases is difficult and usually air flow is reduced.

Safety Measure

- The human body has an electric field and under the right conditions can generate a tremendous amount of voltage, often referred to as static electricity.
- When we open a computer case, we need to make sure that Electro Static Discharge(ESD) does not affect the computers' internal components.

Safety measures

- ESD occurs when an imbalance in the amounts of positive and negative charges on the surface of an object is released.
- The most dramatic example of ESD is lightning, which splits trees as easily as it lights up the sky.
- The amount of energy released when you touch a metal object can be quite large.
- The buildup of energy with nylon clothes (wool is less) can easily reach 21,000 volts.
- About 750 volts are required to produce a visible spark with ESD, while a mere 10 volts or so can ruin a computer chip.

Safety measures

- Sources of ESD are:
 - ▣ The human body has an electric field and it can generate static electricity.
 - ▣ Another source of ESD is dust that build upon pc components. The dust can create charges that could damage internal components of PC.
 - ▣ To avoid this, we have to clean the PC regularly. Blower is the best tool for this.

- The amount of damage caused by ESD can be divided into three categories:
 - 1) **Catastrophic failure:** This damage causes a computer component to fail immediately. Heat and sometimes noise and smoke are generated during the process. This type of failure is easy to identify as the device fail completely. You can replace such components.
 - 2) **Upset failure (degradation failure):** ESD causes erratic fault in a component. This problem is very difficult to detect and repair because the failure is intermittent. The best way to solve this problem is replacing suspected parts until the fault stops appearing.
 - 3) **Latent failure:** this type of failure weakens the actual transistor. The part will seem normal in most operations and will frequently pass quality control and conformance tests. Like upset failure, it is difficult to identify.

Preventing ESD

- Preventing ESD is the best defense against it. The number one cause of ESD is improper handling of electronic devices during installation or maintenance.
- The key prevention to ESD is to keep all electronic components and yourself at a common electric al point. When you open a computer, ground yourself by touching the case (chassis) of a computer.
- There are also devices that enable us to protect ESD. These devices include:
 - 1) **Anti-static wristband:** these devices have a strap that you wrap around your wrist or ankle on one end. The other end of the strap is attached computer case to keep ground between the computer and you.



- 2) **Anti-static mats:** these are mats that you place on the work surface and on the floor in front of work area. They make an excellent place to place your tools and components when you work on a system.



Fig 1.4 Anti-static mat

- 3) **Anti-static bags (pouches):** they have special coating or filament that prevents ESD. They are used to store any sensitive electronic device like cards, RAM, drives, etc when they are not installed in a computer.
- 4) **Anti-static pad:** It is an insulating foam pad in which individual chips with exposed pins should be embedded when they are not installed in a computer.

□ Rules for opening computer:

- 1) Turn off (shut down) the computer
- 2) Remove the power cable (unplug it)
- 3) Remove peripheral devices like mouse, printer, keyboard, etc
- 4) Open the case. Take care of ESD. Wear anti-static wristband or hold the case to ground yourself.
- 5) Take care when you hold internal components of computer like RAM, Cards, CPU, etc. Hold them on the sideways.